

The Grand Architecture of Binding

A Unified Framework Across Scales

This white paper presents a unified mathematical and structural framework connecting atomic shells, gravitational orbital ladders, the universal binding law, the unified collapse operator, and the phenomenon of orbital ionization caused by stellar mass loss.

1. The Universal 1/r Binding Potential

Both atomic and gravitational systems are governed by a central attractive potential of the form:

$$V(r) = -\frac{K}{r}$$

where:

- For gravity: $K = GMm$
- For atoms: $K = k Z_{\text{eff}} e^2$

This shared structure is the foundation of the Grand Architecture: a negative well, a threshold at $E = 0$, and discrete stable radii.

2. Classical Binding Energy

For a circular orbit in a $1/r$ potential:

$$\frac{mv^2}{r} = \frac{K}{r^2} \Rightarrow v^2 = \frac{K}{mr}$$

Total energy:

$$E = \frac{1}{2}mv^2 - \frac{K}{r} = -\frac{K}{2r}$$

This yields the **universal classical binding law**:

$$E_{\text{bound}} = -\frac{K}{2r}$$

3. Quantum Binding Energy

For hydrogen-like atoms:

$$E_n = -\frac{\mu K^2}{2\hbar^2 n^2}$$

This has the same structural form: negative, laddered, and approaching zero from below.

4. The Universal Binding Law

We generalize both classical and quantum forms into:

$$E_{\text{bound}} = -\frac{\mathcal{B}}{\mathcal{S}}$$

where:

- $\mathcal{B}$ = binding strength (gravity: GMm , atom: Z_{eff}^2)
 - $\mathcal{S}$ = scale variable (gravity: r , atom: n^2)
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5. Gravitational Ladders

Planetary orbits follow a geometric ladder:

$$r_k^{(g)} = a_g q^k$$

where $q \approx 1.8\text{--}1.9$ for the outer Solar System.

6. Atomic Shells

Atomic radii follow:

$$r_n^{(a)} = a_a n^2$$

Quadratic spacing instead of geometric, but the same architecture: discrete stable radii in a central potential.

7. Exact Mapping: q-Ladders → n-Shells

Assume a scale conversion:

$$S_g r_k^{(g)} = S_a r_n^{(a)}$$

Substitute the radius laws:

$$S_g a_g q^k = S_a a_a n^2$$

Define:

$$C^2 = \frac{S_g a_g}{S_a a_a}$$

Then the exact mapping is:

$$n = C q^{k/2}$$

and inversely:

$$k = 2 \frac{\ln(n/C)}{\ln q}$$

8. The Unified Collapse Operator

A system becomes unbound when its total energy reaches zero:

$$E = -\frac{\mathcal{B}}{\mathcal{S}} + \Delta E \geq 0$$

Define the collapse operator $\mathcal{C}$:

$$\boxed{\mathcal{C} : (E, \mathcal{B}, \mathcal{S}) \rightarrow \text{unbound if } E + \Delta E \geq 0 \text{ or } \mathcal{B} \downarrow \rightarrow 0}$$

This operator governs:

- Atomic ionization (reducing Z_{eff} or adding photon energy)
 - Orbital ejection (reducing M or adding kinetic energy)
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9. Stellar Mass Loss as Orbital Ionization

Planetary binding energy:

$$E = -\frac{GMm}{2a}$$

If the star loses mass $M \rightarrow M'$:

$$a' = a \frac{M}{M'}$$

and:

$$E' = -\frac{GM'm}{2a'} = -\frac{Gm}{2a} \frac{M'^2}{M}$$

As $M' \downarrow$, $|E'|$ shrinks. When $E' \rightarrow 0$, the planet becomes unbound: **orbital ionization**.

This is directly analogous to atomic ionization: reducing Z_{eff} weakens the Coulomb well until an electron escapes.

10. Conclusion

Across atoms, planets, and galaxies, the same architecture governs binding: a central $1/r$ potential, negative energy wells, discrete stable radii, and a universal collapse operator that determines when systems become unbound.

The mapping between gravitational q-ladders and atomic n-shells reveals a deep structural unity across scales, demonstrating that binding, collapse, and ionization are manifestations of a single underlying law.